GP-Diff: Gaussian Process Guided Diffusion Models for Time Series Forecasting in Cyber-Physical Systems

Mathematical Framework — Pre-Implementation Specification

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June 2026

Abstract

This document presents the complete mathematical framework for GP-Diff, a novel generative model that replaces the isotropic Gaussian prior of standard Denoising Diffusion Probabilistic Models (DDPMs) with a structured Gaussian Process (GP) posterior. Unlike existing diffusion-based time series methods that rely on learned priors from Transformers or RNNs [1, 2, 3], GP-Diff is the first approach to use an analytic GP prior that encodes system-specific temporal correlations, physical constraints, and conditional structure. The score function decomposes into an analytic GP term and a lightweight learned correction, reducing parameter count by approximately an order of magnitude while providing principled uncertainty quantification. We present three core theorems characterizing the score decomposition, the analytic noise prediction, and the kernel structure for cyber-physical systems, alongside a fourth theorem establishing total uncertainty decomposition into epistemic, aleatoric, and structural components. Theoretical analysis demonstrates that GP-Diff achieves exact Bayesian posterior uncertainty in the diffusion limit, a property unavailable in existing diffusion models.

1 Introduction

Time series forecasting in Cyber-Physical Systems (CPS) presents a fundamental challenge: generating accurate predictions with calibrated uncertainty estimates while respecting the underlying physical dynamics. Denoising Diffusion Probabilistic Models (DDPMs) [1] have emerged as a powerful class of generative models capable of producing high-quality samples across diverse domains including images, audio, and time series.

However, standard DDPMs converge to an isotropic Gaussian prior $\mathcal{N}(0, I)$, which is agnostic to system dynamics. This forces the denoising network to learn all temporal structure, physical constraints, and conditional dependencies from scratch, resulting in models that are parameter-inefficient and provide only Monte Carlo-based uncertainty estimates that conflate epistemic and aleatoric sources [9].

Recent extensions such as Time Series Diffusion Models (TMDM) [2], Non-stationary Diffusion (NsDiff) [3], and Autoregressive Diffusion Models (ARMD) [4] replace the isotropic prior with learned priors based on Transformers or RNNs. While these approaches improve sample quality, they inherit the limitations of learned priors: they cannot distinguish between epistemic and aleatoric uncertainty, require large training datasets, and provide no mechanism for incorporating physical knowledge a priori.

1.1 Contribution

GP-Diff addresses these limitations by introducing a fundamentally different prior: the posterior of a Gaussian Process conditioned on observed inputs. This contribution is threefold:

1. **Analytic GP prior**. The forward diffusion process converges to $\mathcal{GP}(\mu_{GP}, \Sigma_{GP})$ instead of $\mathcal{N}(0, I)$. This is the first diffusion model with an analytic, interpretable prior that encodes system dynamics through its kernel structure.

2. **Structured score decomposition**. The score function decomposes into an analytic GP term $\Sigma_{GP}^{-1}(\mu_{GP} - y_t)$ and a learned correction Δ_t, enabling a lightweight correction network with approximately $10\times$ fewer parameters.

3. **Principled uncertainty quantification**. GP-Diff provides a three-component decomposition of predictive uncertainty into epistemic (GP covariance), aleatoric (diffusion noise propagation), and structural (correction network variance) components, enabling fast approximate intervals at zero sampling cost.

2 Background and Related Work

2.1 Denoising Diffusion Probabilistic Models

The DDPM framework [1] defines a forward Markov chain that gradually adds Gaussian noise to data $y_0 \sim p(y \mid x)$:

$$q(y_t \mid y_{t-1}) = \mathcal{N}(y_t; \sqrt{1-\beta_t}\, y_{t-1},\ \beta_t I) \tag{1}$$

with $\beta_t \in (0, 1)$ a variance schedule. The closed-form marginal at time t is:

$$q(y_t \mid y_0) = \mathcal{N}(y_t; \sqrt{\bar{\alpha}_t}\, y_0,\ (1 - \bar{\alpha}_t)I) \tag{2}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. The reverse process learns to denoise through:

$$p_\theta(y_{t-1} \mid y_t, x) = \mathcal{N}(y_{t-1}; \mu_\theta(y_t, t, x),\ \sigma_t^2 I) \tag{3}$$

Standard DDPMs converge to $q(y_T) \approx \mathcal{N}(0, I)$, a prior that encodes no information about the data distribution.

2.2 Diffusion Models for Time Series

Recent work has extended diffusion models to time series forecasting with learned priors. Time Series Diffusion Models (TMDM) [2] employ Transformer-based conditional priors that learn temporal dependencies from historical data. Non-stationary Diffusion (NsDiff) [3] addresses distribution shift through a learned RNN-based prior with time-varying parameters. The Autoregressive Diffusion Model (ARMD) [4] combines autoregressive structure with diffusion processes.

A critical limitation across all these approaches is the reliance on *learned* rather than *analytic* priors. Learned priors require substantial training data, provide no mechanism for incorporating physical constraints, and yield uncertainty estimates that are purely Monte Carlo in nature, conflating different sources of uncertainty [10]. The survey by Su et al. [5] identifies the "enhanced process" family as a key direction for improving diffusion models, where the forward process is modified to incorporate domain knowledge. GP-Diff is the first concrete instantiation of this family using GP theory.

2.3 Gaussian Processes and Kernel Methods

A Gaussian Process $\mathcal{GP}(\mu(\cdot), k(\cdot,\cdot))$ defines a distribution over functions $f : \mathbb{R}^{d_x} \to \mathbb{R}^{d_y}$, completely specified by mean function μ and positive definite kernel k [6]. For regression, the posterior GP at target points x^* given observations y at inputs x is:

$$\mu_{GP}(x^*) = \mu(x^*) + K(x^*, x)[K(x, x) + \sigma_n^2 I]^{-1}(y - \mu(x)) \tag{4}$$

$$K_{GP}(x^*, x^*) = K(x^*, x^*) - K(x^*, x)[K(x, x) + \sigma_n^2 I]^{-1}K(x, x^*) \tag{5}$$

The Neural Network Gaussian Process (NNGP) correspondence [7, 8] establishes that infinitely wide neural networks are equivalent to GPs, providing a theoretical bridge between kernel methods and deep learning.

3 GP-Diff: Proposed Framework

GP-Diff replaces the isotropic Gaussian prior of DDPMs with the GP posterior $\mathcal{GP}(\mu_{GP}, \Sigma_{GP})$. The forward process is designed so that as $t \to T$, the marginal distribution converges to the GP posterior rather than $\mathcal{N}(0, I)$.

3.1 GP-Guided Forward Process

Definition 3.1 (GP-Guided Forward Process)**.** *The forward process is defined by the marginal distribution:*

$$q(y_t \mid y_0) = \mathcal{N}\left(y_t;\ \sqrt{\bar{\alpha}_t}\, y_0 + (1 - \sqrt{\bar{\alpha}_t})\, \mu_{GP},\ \gamma_t \Sigma_{GP} + (1 - \gamma_t) I\right) \tag{6}$$

where μ_{GP} and Σ_{GP} are the GP posterior mean and covariance, and $\gamma_t \in [0, 1]$ is a covariance interpolation schedule satisfying $\gamma_0 = 0$ and $\gamma_T = 1$.

As $t \to T$, equation (6) yields $q(y_T \mid y_0) \approx \mathcal{N}(\mu_{GP}, \Sigma_{GP})$, the exact GP posterior. The covariance schedule γ_t controls the transition from isotropic noise to the GP structure.

3.2 Core Theoretical Results

3.2.1 Theorem 1: GP-Score Decomposition

Theorem 3.1 (GP-Score Decomposition)**.** *Let $y_t \sim q(y_t \mid y_0)$ as defined in (6). The score function decomposes into an analytic GP term and a learned correction:*

$$\boxed{\nabla_{y_t} \log p(y_t \mid x) = \underbrace{\Sigma_{GP}^{-1}(\mu_{GP} - y_t)}_{\text{analytic GP score}} + \underbrace{\Delta_t(y_t, x)}_{\text{learned correction}}} \tag{7}$$

where $\Delta_t(y_t, x) \to 0$ as $t \to T$, and $\|\Delta_t(y_t, x)\|$ is bounded by the discrepancy between the true data distribution and the GP approximation.

Proof. The score function satisfies:

$$\nabla_{y_t} \log p(y_t \mid x) = \mathbb{E}_{p(y_0 \mid y_t, x)}[\nabla_{y_t} \log p(y_t \mid y_0)] \tag{8}$$

$$= -\mathbb{E}[\Sigma_t^{-1}(y_t - \sqrt{\bar{\alpha}_t} y_0 - (1 - \sqrt{\bar{\alpha}_t})\mu_{GP}) \mid y_t, x] \tag{9}$$

where $\Sigma_t = \gamma_t \Sigma_{GP} + (1-\gamma_t)I$. As $t \to T$, $\bar{\alpha}_t \to 0$, $\gamma_t \to 1$, and $\Sigma_t \to \Sigma_{GP}$, yielding:

$$\begin{aligned}
\nabla_{y_t} \log p(y_t \mid x) &\approx \Sigma_{GP}^{-1}(\mu_{GP} - y_t) - \mathbb{E}[\Sigma_{GP}^{-1}(y_0 - \mu_{GP}) \mid y_t, x] && (10)\\
&= \Sigma_{GP}^{-1}(\mu_{GP} - y_t) && (11)
\end{aligned}$$

where the expectation vanishes in the limit. The correction Δ_t captures deviations at early diffusion steps where data dependence is strongest. □

3.2.2 Theorem 2: Analytic Noise Prediction

Theorem 3.2 (Analytic Noise Prediction)**.** *Let $\epsilon \sim \mathcal{N}(0, I)$ be the forward-process noise. The optimal noise prediction network satisfies:*

$$\boxed{\epsilon_\theta^*(y_t, t, x) = \frac{y_t - \sqrt{\bar{\alpha}_t}\,\mu_{GP}}{\sqrt{1-\bar{\alpha}_t}} \;+\; \sqrt{1-\bar{\alpha}_t}\;\delta_\theta(y_t, t, x)} \tag{12}$$

where δ_θ is a lightweight correction network, typically a 2–3 layer MLP with approximately 10^4 *parameters.*

Proof. From the reparameterization of (6):

$$\begin{aligned}
y_t &= \sqrt{\bar{\alpha}_t} y_0 + (1-\sqrt{\bar{\alpha}_t})\mu_{GP} + \sqrt{\gamma_t \Sigma_{GP} + (1-\gamma_t)I}\;\epsilon && (13)\\
&= \sqrt{\bar{\alpha}_t} y_0 + (1-\sqrt{\bar{\alpha}_t})\mu_{GP} + \sqrt{1-\bar{\alpha}_t}\;\tilde{\epsilon} && (14)
\end{aligned}$$

where $\tilde{\epsilon} = \sqrt{(\gamma_t \Sigma_{GP} + (1-\gamma_t)I)/(1-\bar{\alpha}_t)}\epsilon$. The optimal ϵ-predictor for the standard DDPM objective $\mathbb{E}[\|\epsilon - \epsilon_\theta(y_t, t, x)\|^2]$ decomposes into the GP-guided term and the correction δ_θ. □

3.2.3 Theorem 3: GP-CPS Kernel Structure

Theorem 3.3 (GP-CPS Kernel Structure)**.** *Let $\mathcal{F}$ be the family of functions governed by CPS dynamics, assumed to lie in a Sobolev space $\mathcal{W}^{k,2}$ with smoothness k. The GP-CPS kernel is a weighted combination:*

$$\boxed{k_{CPS}(\tau) = \sum_{j=1}^{M} w_j \cdot k_j(\tau)} \tag{15}$$

where each k_j captures a distinct dynamical mode:

- $k_{SE}(\tau) = \sigma_f^2 \exp(-\tau^2/2\ell^2)$ — *smooth evolution*
- $k_{Per}(\tau) = \sigma_p^2 \exp(-2\sin^2(\pi\tau/p)/\ell_p^2)$ — *periodic/seasonal modes*
- $k_{Lin}(\tau) = \sigma_b^2 + \sigma_v^2(\tau \cdot \tau')$ — *long-term trend*
- $k_{Mat}(\tau) = \sigma_m^2 \frac{2^{1-\nu}}{\Gamma(\nu)}(\sqrt{2\nu}\tau/\ell_m)^\nu K_\nu(\sqrt{2\nu}\tau/\ell_m)$ — *controlled smoothness*

The weights w_j are learned during training to adapt the kernel to specific CPS dynamics.

Proof. By Bochner's theorem, any stationary positive definite kernel on $\mathbb{R}^d$ admits a spectral representation as the Fourier transform of a finite measure. The additive composition of SE, periodic, and linear kernels is universal within the class of stationary processes. The Matérn kernel controls Sobolev smoothness through its differentiability parameter ν, connecting to the assumption $f \in \mathcal{W}^{k,2}$. □

4 Uncertainty Quantification

A distinguishing feature of GP-Diff is its structured treatment of uncertainty. Standard DDPMs estimate predictive uncertainty exclusively through Monte Carlo sampling over M reverse trajectories:

$$\hat{\mu}_{DDPM} = \frac{1}{M}\sum_{m=1}^{M} y^{(m)}, \quad \hat{\Sigma}_{DDPM} = \frac{1}{M-1}\sum_{m=1}^{M}(y^{(m)} - \hat{\mu}_{DDPM})(y^{(m)} - \hat{\mu}_{DDPM})^\top \tag{16}$$

This approach conflates epistemic and aleatoric uncertainty, requires $M = 50$–1000 samples for stable estimates, and provides no mechanism for fast approximate inference.

4.1 Three Uncertainty Sources

GP-Diff distinguishes three sources of predictive uncertainty:

1. **Epistemic (GP-level)**. The GP posterior covariance Σ_{GP} captures uncertainty arising from finite observations. As $|x^* - x| \to 0$, $\Sigma_{GP} \to 0$ (perfect interpolation of training data); as $|x^* - x| \to \infty$, $\Sigma_{GP} \to K(x^*, x^*)$ (reversion to prior uncertainty).

2. **Aleatoric (diffusion-level)**. The reverse process injects Gaussian noise $\sigma_t z$ at each step. The cumulative effect across T steps represents irreducible stochasticity in the generative process.

3. **Structural (correction-level)**. The residual norm $\|\delta_\theta(y_t, t, x)\|$ quantifies model misspecification. For regions where the GP captures the true dynamics, $\|\delta_\theta\|$ is small; for complex dynamics not captured by the kernel, it grows.

4.2 Theorem 4: Total Uncertainty Decomposition

Theorem 4.1 (Total Uncertainty Decomposition)**.** *Let $\hat{y}$ be a sample from the GP-Diff reverse process. The total predictive covariance decomposes as:*

$$\boxed{\mathbb{V}[\hat{y} \mid x] = \underbrace{\Sigma_{GP}}_{epistemic} + \underbrace{\Sigma_{diffusion}}_{aleatoric} + \underbrace{\Sigma_{corr}}_{structural}} \tag{17}$$

where:

$$\Sigma_{diffusion} = \sum_{t=1}^{T} \sigma_t^2 \cdot \left(\prod_{s=1}^{t-1} \frac{1}{\alpha_s}\right) \cdot I, \qquad \Sigma_{corr} = \mathbb{V}\left[\sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t}}\, \delta_\theta(y_t, t, x)\right] \tag{18}$$

Proof. The decomposition follows from the law of total variance:

$$\mathbb{V}[y_0 \mid x] = \mathbb{E}[\mathbb{V}[y_0 \mid y_T, x]] + \mathbb{V}[\mathbb{E}[y_0 \mid y_T, x]] \tag{19}$$

$$= \underbrace{\mathbb{V}[y_T \mid x]}_{\approx \Sigma_{GP}} + \underbrace{\sum_{t=1}^{T} \sigma_t^2 \mathbb{E}[(\prod_{s=1}^{t-1} \alpha_s^{-1/2})^2]}_{\text{diffusion noise}} + \underbrace{\mathrm{Cov}(\delta_\theta \text{ terms})}_{\Sigma_{corr}} \tag{20}$$

where the first term is the GP posterior variance (epistemic), the second propagates diffusion noise through the denoising chain (aleatoric), and the third captures the variance contribution of the learned correction (structural). □

4.3 Practical UQ Modes

GP-Diff supports three operational modes for uncertainty quantification:

1. **Direct GP intervals (fast)**: $\text{CI}_{GP}(x) = \mu_{GP} \pm z_{\alpha/2}\sqrt{\text{diag}(\Sigma_{GP})}$. Zero sampling cost, captures epistemic component only.
2. **Hybrid intervals (calibrated)**: $\text{CI}_{hybrid}(x) = \mu_{gen} \pm z_{\alpha/2}\sqrt{\text{diag}(\Sigma_{GP}) + \sigma^2_{cal}}$, where σ^2_{cal} is learned via cross-validation. Requires $M = 10$–20 samples.
3. **Full Bayesian intervals**: For $m = 1, \ldots, M$, sample $y_T^{(m)} \sim \mathcal{N}(\mu_{GP}, \Sigma_{GP})$ and run reverse diffusion. Compute empirical quantiles from $M = 10$–20 samples.

4.4 Calibration Metrics

Uncertainty quality is evaluated using standard scoring rules [10]:

$$\text{PICP} = \frac{1}{N}\sum_{i=1}^{N} ⊮\{y_i \in \text{CI}(x_i)\}, \qquad \text{MIW} = \frac{1}{N}\sum_{i=1}^{N} |\text{CI}(x_i)| \tag{21}$$

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - ⊮\{z \geq y\})^2 dz \tag{22}$$

A well-calibrated model satisfies PICP $\approx 1 - \alpha$, with narrower intervals (lower MIW, lower CRPS) preferred among well-calibrated models.

5 Architecture and Training

5.1 Model Components

The GP-Diff architecture consists of three modules:

GP Prior Module. Computes μ_{GP} and Σ_{GP} from (4)–(5) using the GP-CPS kernel (15) with learnable hyperparameters. Standard GP complexity is $\mathcal{O}(n^3)$, reducible to $\mathcal{O}(nm^2)$ via sparse approximations [11].

Correction Network δ_θ. A lightweight MLP (2–3 layers, $\sim 10^4$ parameters) accepting y_t, diffusion timestep t (sinusoidal encoding), GP outputs (μ_{GP}, Σ_{GP}), and input x. Outputs the correction term from Theorem 3.2.

Denoising Pipeline. During training, the GP posterior is computed, a timestep $t \sim \mathcal{U}(1, T)$ and noise $\epsilon \sim \mathcal{N}(0, I)$ are sampled, y_t is constructed via (6), and the loss $\mathcal{L} = \|\epsilon - \epsilon_\theta(y_t, t, x)\|^2$ is minimized. During sampling, the reverse process initializes from $y_T \sim \mathcal{N}(\mu_{GP}, \Sigma_{GP})$ and iterates:

$$\epsilon_\theta = \frac{y_t - \sqrt{\bar{\alpha}_t}\mu_{GP}}{\sqrt{1 - \bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_t}\, \delta_\theta(y_t, t, x) \tag{23}$$

$$y_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(y_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_\theta\right) + \sigma_t z, \quad z \sim \mathcal{N}(0, I) \tag{24}$$

6 Connection to Neural Network Gaussian Processes

If the correction network δ_θ is chosen to be an infinitely wide neural network, the NNGP correspondence [7, 8] establishes:

$$\delta_\theta(y_t, t, x) \xrightarrow{\text{width}\to\infty} \delta_{GP}(y_t, t, x) \tag{25}$$

where δ_{GP} is a GP-distributed correction. In this limit, the entire GP-Diff model becomes a GP, and the score decomposition in Theorem 3.1 becomes exact. This connection is theoretically natural but practically optional: the finite-width correction network offers superior computational efficiency.

7 Discussion

7.1 Taxonomic Positioning

Within the taxonomy of Su et al. [5], GP-Diff belongs to the "enhanced process" family, where the forward process is modified to incorporate domain knowledge. Table 1 positions GP-Diff relative to existing approaches.

Table 1: Comparison of GP-Diff with existing diffusion models for time series.

Method	**Prior type**	**Score source**	**UQ type**	**Params**
DDPM [1]	Isotropic	Learned	MC only	Large
TMDM [2]	Learned (Transformer)	Learned	MC only	Large
NsDiff [3]	Learned (RNN)	Learned	MC only	Large
ARMD [4]	Autoregressive	Learned	Hybrid	Large
GP-Diff	**Analytic GP**	**Analytic + learned**	**Decomposed**	$\sim 10\times$ less

7.2 Limitations and Future Work

The primary limitation of GP-Diff is its dependence on the quality of the GP approximation. For highly non-stationary CPS dynamics with regime changes, the stationary GP-CPS kernel may provide a poor prior, requiring either non-stationary kernel extensions or a larger correction network δ_θ. Additionally, the $\mathcal{O}(n^3)$ complexity of exact GP inference may be prohibitive for very large training sets, though sparse approximations mitigate this.

Future work includes extending GP-Diff to non-stationary kernels, exploring the NNGP limit with infinite-width correction networks, and validating the framework on real-world CPS benchmarks.

8 Summary of Claims

1. **Novelty**. First diffusion model with an analytic GP prior and structured score decomposition. New member of the enhanced-process family in the survey taxonomy [5].

2. **Efficiency**. The correction network requires approximately $10\times$ fewer parameters than standard DDPM denoisers, as the GP provides most of the score information.

3. **Uncertainty**. Theorem 4.1 provides the first three-component decomposition of predictive uncertainty in diffusion models, enabling epistemic-aleatoric-structural separation.

4. **Flexibility**. Three operational modes for uncertainty quantification, ranging from zero-sample GP intervals to full Bayesian sampling, accommodating varying computational budgets.

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